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RESEARCH ARTICLE

An Affordable AI-Driven and 3D-Printed Personalized Myoelectric Prosthesis: Design, Development, and Assessment

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ABSTRACT Upper-limb amputations significantly affect independence and quality of life, particularly in low-income regions where advanced prosthetic technology is costly and lacks adequate personalization. Conventional myoelectric prostheses, while offering functional restoration, have limited adaptability and high cost. This study presents a personalized transradial myoelectric prosthesis that combines additive manufacturing and Artificial Intelligence (AI) control, offering an accessible and high-performance solution. The prosthesis design utilizes additive manufacturing (3D printing) for anatomical personalization via 3D scanning and parametric modeling. An AI-driven control system utilizes machine learning to classify electromyography (EMG) signals in real-time, specifically detecting the user's intention to perform flexion or extension movements, and tailoring responses to individual users. Evaluation employed the "Brief Activity Measure for Upper Limb Amputees (BAM-ULA)" protocol with nine participants with transradial amputations. Trials with the nine participants yielded an average BAM-ULA score of 7.4 out of 10 (Standard Deviation (SD) 0.7). This demonstrated robust functional performance, comparable to high-end commercial devices in initial tests. Gross motor tasks saw 100% success rates; fine motor tasks, 22.2%. Integrating AI and additive manufacturing resulted in an affordable, high-performance, personalized prosthesis. This work highlights how localized digital manufacturing enables accessible customization for users in low-resource settings. The main novelty is this validated integration of personalized additive manufacturing and adaptive AI control in an affordable transradial prosthesis addressing the needs of developing countries.

INDEX TERMS Additive manufacturing, artificial intelligence, BAM-ULA protocol, myoelectric prosthesis, personalized prosthetics.

I. INTRODUCTION

An estimated 3 million people worldwide live with upper limb amputations [1], a condition that profoundly impacts both physical functionality [2] and psychological well-being [3]. Upper limb loss significantly impairs daily

activities [4], limiting independence and overall quality of life [5]. Approximately four out of five of these individuals reside in low-income countries [6], where access to affordable prosthetic care remains limited, exacerbating the challenges they face [7].

In this context, myoelectric prosthetics (MEP) have emerged as an advanced technological solution that allows users to partially regain the functionality of their limbs

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through detecting and processing electromyographic (EMG) signals generated by residual muscles [8]. Recent research has focused on improving the functionality, accessibility, and affordability [9] of MEPs through innovative designs and technologies. For instance, the Galileo Hand, developed by Fajardo et al. [10], provided an affordable solution with a cost under \$350 and a weight of less than 350 g. Its subactuated mechanisms enabled six thumb movements, allowing versatile grip patterns and finger motions. Similarly, Ccorimanya et al. [11] introduced a lightweight (93.3 g) prosthesis with single-channel EMG control, child-friendly auditory feedback, and rapid response times. However, the reliance on 3D scanning for customization posed significant scalability challenges as the child grew, and a single-participant evaluation limited the study's findings.

Other innovations have explored alternative sensing technologies, such as the work by Prakash et al. [12], which integrated a force myography sensor into a 3D-printed prosthetic. This sensor, designed with a unique mechanical assembly and signal conditioning circuitry, demonstrated high sensitivity, repeatability, low hysteresis, and excellent frequency response. These advancements underscore the potential for cost-effective, high-functional prosthetic solutions while highlighting ongoing scalability and broad adoption challenges.

Despite these advancements, significant challenges persist in two key areas: accessibility in low-income regions and adaptability to individual users. The high cost of commercial myoelectric prostheses continues to hinder widespread adoption [13], limiting access for those who could benefit most [14]. Furthermore, the customization options in existing solutions remain insufficient, affecting user comfort, functionality, and long-term acceptance [15]. Many prosthetic designs rely on fixed configurations that do not fully accommodate anatomical and functional variations between individuals, reducing usability and control accuracy. This lack of personalization can result in discomfort [16], poor weight distribution [17], and a limited range of motion [18], ultimately impacting the user's experience and willingness to adopt the device.

In addition to cost and customization barriers, another critical limitation lies in the control mechanisms employed by current myoelectric prostheses. While some devices have explored alternative sensing approaches, such as force myography [19] or low-cost electromyographic models [20], [21], [22], the integration of artificial intelligence for real-time adaptive control remains underdeveloped. Most commercially available prostheses rely on pre-set algorithms that fail to dynamically adjust to variations in muscle activation patterns [23], leading to inconsistent gesture recognition, high false activation rates, and a lack of adaptability across different users [24], [25]. These limitations underscore the need for a more sophisticated control approach that enhances accuracy while maintaining efficiency and usability.

Addressing these issues, this paper presents the design, development, and validation of a personalized transradial myoelectric prosthesis incorporating an AI-based control system. The proposed approach aims to overcome current technological limitations and establishes a method for generating accessible and customized prostheses designed explicitly for implementation in developing countries. The proposed prosthesis combines additive manufacturing technologies for anatomical customization with deep learning algorithms for precise movement control. Additionally, a comprehensive evaluation of the system is conducted in terms of performance, usability, and user acceptance, using the BAM-ULA evaluation protocol as a key tool to measure the impact of personalization. Results from trials with nine participants demonstrated an average BAM-ULA score of 7.4 out of 10, confirming the viability of the proposed approach in providing a competitive and cost-effective prosthetic solution.

The main contributions of this work include (1) the development of an adaptive control system based on AI for myoelectric prostheses, (2) the integration of additive manufacturing technologies to ensure optimal customization, and (3) the experimental validation of the system through user testing in low-income countries. This holistic approach seeks to advance the state-of-the-art in myoelectric prostheses and provide a more effective and accessible solution for individuals with transradial amputations.

II. MATERIALS AND METHODS

This section describes the development of the proposed prosthetic system, dividing the methodology into two key aspects: hardware, which includes the physical components, mechanical design, and fabrication methods, and software, which details the AI-based control system and signal processing. Additionally, the BAM-ULA assessment protocol, which was used to evaluate the prosthesis's functionality, is presented.

A. HARDWARE

This section details the physical components of the proposed prosthetic system, focusing on its mechanical design, fabrication process, and integration of electronic elements. The prosthesis was developed using a user-centered approach. The following subsections outline the design methodology, component selection, and assembly process. A macro diagram is presented in Fig. 1, illustrating the parallel process of designing the parametric hand and prosthetic socket designs.

1) 3D SCANNING AND MEASURES

Hand measurements were conducted using a soft tape measure and a vernier caliper. The parameters collected included the total length of each finger, the palm's width and height, the total distance from the tip of the middle finger to the wrist, and the perimeter and reference diameter of the wrist. Forearm measurements were also taken, including the distance from

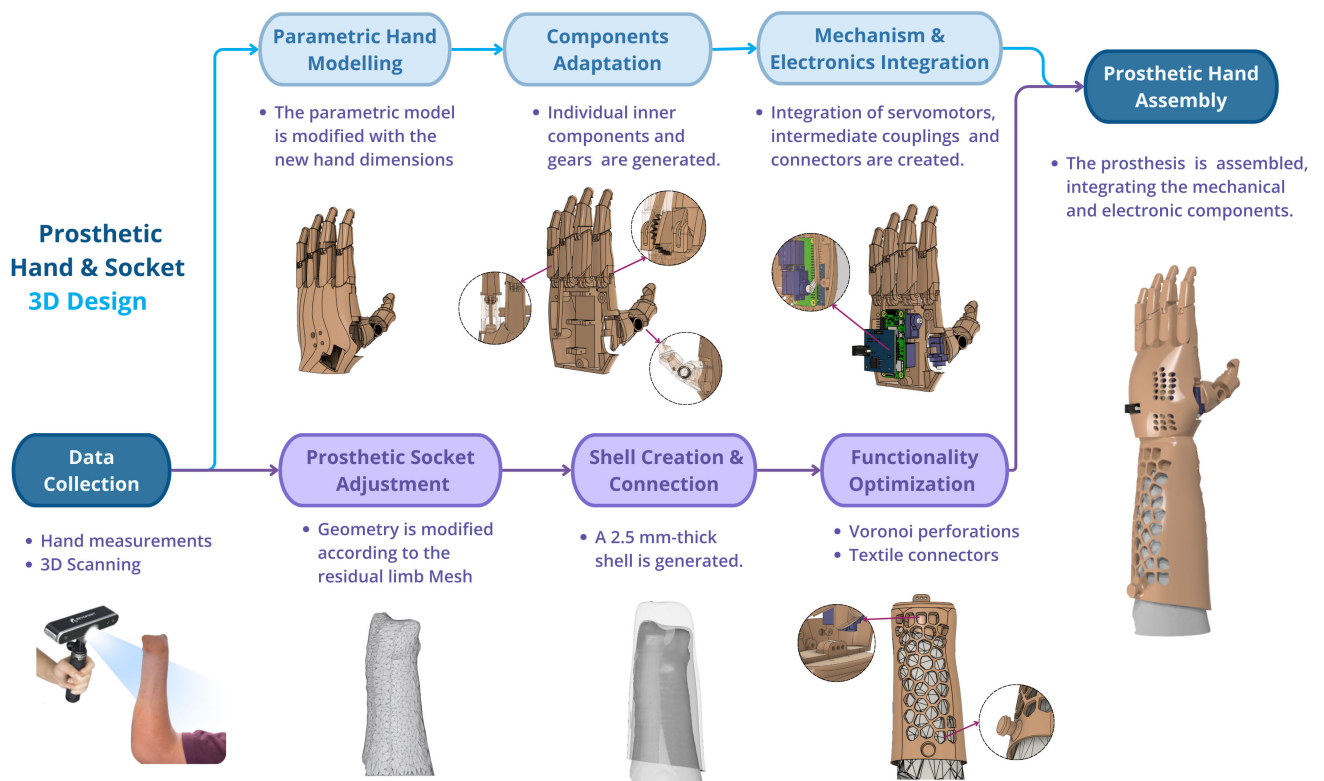


FIGURE 1. 3D Design process of the parametric prosthetic hand and the personalized prosthetic socket.

the wrist to the elbow, the most significant reference diameter of the forearm, and its maximum perimeter. Additionally, a tracing of the hand and the residual limb was made on graph paper, emphasizing the location of the wrist flexion area and the position of the pisiform bone protruding from the wrist.

Next, the residual limb was digitized using a POP 2 3D scanner (Revopoint) in conjunction with the Revo Scan application. For scanning, the depth cameras were configured with an exposure above 6 (on a scale of 1 to 10), aided by the phone's flash if ambient lighting was insufficient. In contrast, the "RGB Camera" used automatic exposure to adapt to ambient conditions. During the scan, the distance between the scanner and the residual limb was maintained in the optimal range suggested by the software, and the data was recorded continuously. The fusion of the resulting point cloud was performed using the advanced method with 15% isolation and default overlap detection and smoothing parameters. Subsequently, a mesh was generated by enabling automatic hole-filling (with default ratio parameters). The resulting .ply file was post-processed in Meshmixer to correct model orientation, close remaining holes, fix overlapping surfaces, smooth textures, and remove extraneous geometry. Finally, the file was exported in .obj format.

2) PARAMETRIC HAND DESIGN

Retopology was applied to the .obj file to prepare the 3D model for computer-aided design (CAD) using Autodesk

Fusion 360. In this process, the original mesh served as a reference to create a new object using the "Freeform" tool in Fusion 360. Specifically, a T-Spline cylinder with at least eight height faces and ten diameter faces was fitted around the reference mesh, and its top surface was closed using the "Fill Hole" tool set to Collapse mode. The resulting T-Spline was then converted to a BRep model, generating a surface that could be exported as a .ipt file for subsequent editing in Autodesk Inventor.

During the design phase, the parametric method proposed by Bustamante et al. [26] was implemented using an editable project file (Parametric_Left_Hand.ipj), associated .ipt files, an Excel file (Parametric_Measurements_Hand.xlsx), and an .exe file that yields optimized parameters. The overall design of the hand was determined based on measured values of finger length, hand width, palm length, and wrist diameter. These dimensions were inputs to generate intermediate parameters for the phalanges and other mechanical features. The Inventor project (.ipj) and corresponding parts (distal phalanx, proximal phalanx, phalanx bar for each finger, two-part palm, wrist connector, wrist, and the wrist's internal pin) were updated accordingly. Since the user-specific wrist components were not required, the connectors linked to these parts were removed. To ensure functional fabrication via 3D printing, interference checks were performed using Inventor's "Analyze Interference" tool, and hole diameters and wall thicknesses were adjusted where necessary. The total lengths of the complete hand and individual fingers were further

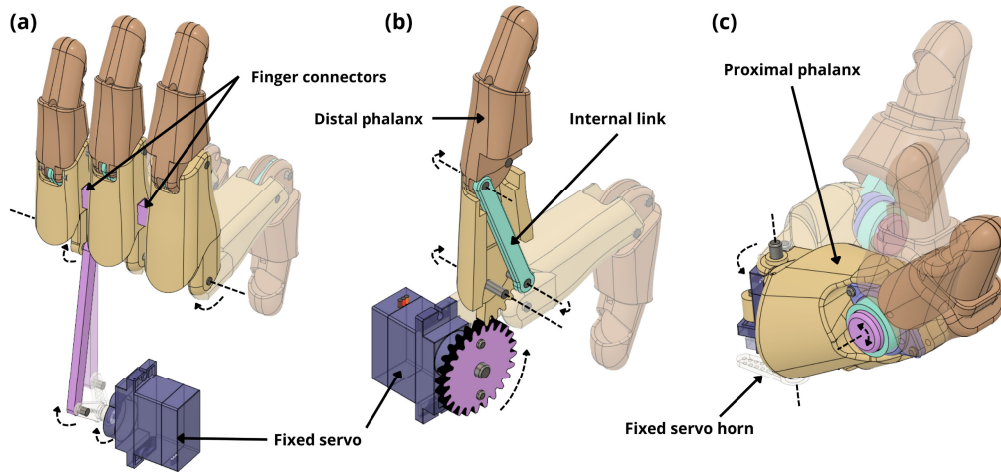


FIGURE 2. Fingers actuation mechanisms. (a) fingers group mechanism, (b) index mechanism, and (c) thumb mechanism.

validated, and if discrepancies were found, the parametric Excel file was revised.

Following the prosthetic hand design described by Cifuentes-Cuadros et al. [27], the two palm components were converted into shells using the “Shell” operation. Internal cavities were created to house two B13DLM V2 HV servomotors and one EMAX ES08MD II motor in the thumb region. The positions and angles of each finger were corrected to align their pivot points and ensure that all finger bases lie at the same height. Additionally, a cut was made in the palm to avoid restricting the thumb’s range of motion. Electronic components (control board and voltage regulator) were secured within the palm using specially designed support posts; power connectors and ventilation openings were also integrated.

Depending on whether the left or right hand was being manufactured and on the user’s anatomical dimensions, the motors, mechanical linkages, and electronic components were positioned according to the measured geometry. All actuators are embedded within the palm structure, thus classifying the prosthesis as an extrinsically actuated system. Finally, an internal cavity at the base of the palm was added to accommodate the prosthetic socket.

Finger actuation is achieved as follows: the middle (Digit III), ring (Digit IV), and little (Digit V) fingers move together as a group through a bar mechanism linked to the B13DLM servomotor at their proximal phalanges, while the index finger (Digit II) is actuated by a gear transmission connecting the its own B13DLM servomotor to its proximal phalanx. The thumb (Digit I) mechanism adapts the HACKberry design by Exiii Inc. to the parametric structure proposed by Bustamante et al. [26], enabling active adduction and abduction movements using the EMAX ES08MD II servomotor, with the proximal and distal phalanx lengths customized to the user’s specific measurements and a passive mechanism that enables the flexion and extension of the thumb.



FIGURE 3. Prosthetic hand gestures.

After integrating all the internal mechanisms, a second interference check was performed at extreme positions for each digit or group of digits, incorporating internal 2mm stainless steel shafts within the rotational joints. Further modifications to palm width, post locations, or component placement were made to achieve a functional prosthetic hand faithful to the user’s anatomical dimensions if necessary.

With this design finalized, the prosthetic hand features three active degrees of freedom (DOF): one for the adduction/abduction of the thumb (90° range), one for the flexion/extension of the index (90° range), and one for the flexion/extension of the group comprising the middle, ring, and pinky fingers (90° range); as well as one passive DOF for the thumb’s flexion and extension movement (30° range), as shown in Fig. 2. The hand can perform six predefined

gestures involving passive and active movement phases, which are shown in Fig. 3.

- **Default:** The hand remains relaxed; in an anatomically neutral position.
- **Pinch:** A precision grip is formed between the index finger and thumb.
- **Cylinder 1:** The fingers fully close together until the index finger makes contact with the thumb, enabling a firm cylindrical grasp.
- **Cylinder 2:** A partial version of *Cylinder 1*, where the fingers close to approximately 50% of their full range.
- **Handle:** The thumb is positioned in supination while the index, middle, ring, and little fingers fully close, allowing a secure grasp of larger objects.
- **Typing:** The thumb is supinated, the index finger remains extended, and the middle, ring, and little fingers are fully flexed, enabling fine motor interaction such as typing on a keyboard.

3) PROSTHETIC SOCKET DESIGN

The prosthetic socket was designed based on the 3D model scan of the user’s residual limb. If the limb had a conical shape, no significant adjustments were needed. However, if any narrower sections were present along the forearm, the geometry was modified to ensure smooth insertion and removal of the limb, preventing it from getting stuck.

Once the final limb shape was established, a 2.5 mm-thick shell was generated to serve as the socket’s base structure. The top section, which connects to the prosthetic hand, was then designed. This involved aligning the socket with the palm components and making a 2.3 cm cut at the base of the palm to create a seamless transition between the hand and socket. A mounting piece was added to secure the palm to the socket using screws.

Voronoi perforations were incorporated into the socket to improve breathability, comfort, and mechanical performance. As demonstrated by Rodriguez et al. [28], this pattern helps distribute forces more evenly, reduces stress concentration, and increases ventilation, making the prosthesis lighter and more comfortable to wear. Finally, two connectors were integrated into the socket to attach a textile forearm support, ensuring a secure fit of the prosthetic system to the user.

Each socket features a distal universal connector, ensuring secure interfacing with the developed prosthetic hands.

The final step is to ensure the correct fitting of the superficial EMG (sEMG) sensors to complete the prosthetic socket. The donning process began with placing sEMG electrodes on the forearm and upper arm. A prosthetic sleeve was applied over the forearm, then the residual limb was inserted into the prosthetic socket. The prosthesis was secured using an adapted elbow brace positioned over the electrodes to prevent unwanted slippage. This brace featured two button-like openings interlocked with two equidistant

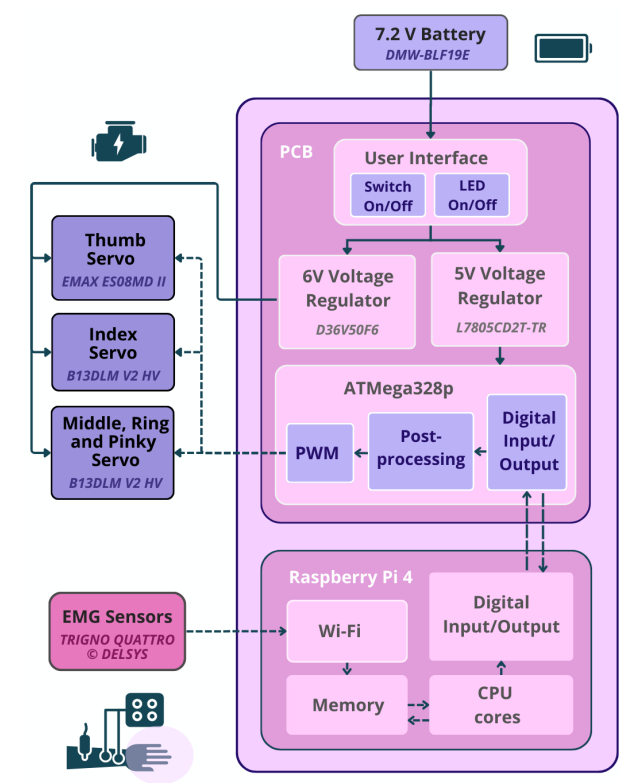


FIGURE 4. Embedded system architecture for the prosthetic hand controller with EMG-based gesture recognition.

TABLE 1. Technical Specifications of the Trigno® Quattro Sensor 2024 by Delsys Incorporated.

Feature	Specification
EMG Channels	6 channels
Communication	Wireless
Filtering	Butterworth band-pass filter (40–80 dB/dec), RMS envelope (100 ms window)
Amplification	CMRR > 80 dB, input range ± 11 mV or ± 22 mV, 16-bit resolution
Transmission Range	Up to 40 meters in RF mode
Wireless Protocols	ISM band (2.400–2.483 GHz), proprietary RF protocol, BLE 4.2
Sensor Size	27×46×13 mm; sensor head: 25×12×7 mm
Weight	25 g
Battery Life	2–4 hours
Material	Medical-grade polycarbonate, UV-sealed for moisture protection

protruding pins on the proximal end of the prosthetic socket.

4) ELECTRONIC DESIGN

The electronic system is composed of two primary elements: a Raspberry Pi Zero 2W and a custom-printed circuit board (PCB) developed specifically for this project, which incorporates an ATmega328P microcontroller. The Raspberry Pi handles the acquisition and processing of EMG signals through an AI-based algorithm. It establishes a wireless connection with two Trigno® Quattro Sensor 2024 multichannel EMG sensors (Delsys Inc.), which capture, preprocess, and store EMG data in their internal memory. A complete diagram

of the system's architecture is provided in Fig. 4, while Table 1 outlines the sensors' key technical specifications [29].

Communication between the sensor and the Raspberry Pi module is initiated by an activation command sent from the Raspberry Pi. Subsequently, the sensors acquire signals from all channels, perform onboard preprocessing, including filtering to mitigate electromagnetic interference, and store the resulting voltage data in their internal memory. The Raspberry Pi receives the data from the sensors in packets containing five samples per channel, a format chosen to balance real-time processing demands with communication latency. This information is stored in a specific data structure that is constantly refreshed to support the AI algorithm's processing needs. Meanwhile, the ATmega328P handles post-processing of the AI output and controls the servomotors.

Each Trigno sensor employs three electrodes, resulting in a total of six channels. The electrodes are placed equidistantly around the forearm—three over flexor muscles and three over extensor muscles, near the elbow. A reference electrode is placed on the upper arm, while all other electrodes are secured using double-sided adhesive pads to ensure stability and comfort for the user. The system requires no calibration and exhibits consistent performance across multiple sessions.

This setup is designed for individuals with distal transradial amputations and has been validated in adults of varying forearm sizes. Reliable EMG acquisition was observed during testing by matching electrode placement to the configuration used during training data acquisition. The user's skin is cleaned with alcohol at the electrode placement sites, avoiding dense hair or scar tissue, as part of the testing procedure. Depending on the battery level, the system operates continuously for 2-4 hours, and adhesive performance remains stable during one-hour test sessions. Prosthetic socket integration details are presented in a subsequent section.

The PCB also features a 5V voltage regulator to power the Raspberry Pi and the ATmega328P. This is achieved using an L7805CD2T-TR linear voltage regulator. To support the actuation system, which requires 6V for the servomotors, the design incorporates a D36V50F6 step-down regulator, ensuring stable voltage and sufficient current supply to all three servomotors during operation.

Several design considerations were made, the most critical being the available space. Given the embedded nature of the system, the workspace is constrained to 5 cm × 7 cm × 3 cm for the electronic components. The trace widths were carefully calculated when designing the PCB based on the expected current draw. For the 5 V traces, the estimated maximum current is 1 A, resulting in a trace width of 0.5 mm, which can support up to 1.5 A. For the 6 V traces, which handle the current for all three servomotors operating at full load, the maximum estimated current is 2.7 A. A trace width of 1 mm was chosen to accommodate this, capable of handling up to 2.8 A [30].

TABLE 2. Manufacturing parameters for different prosthetic components.

Feature	Mechanism activation parts	Protective covers parts
Layer height (mm)	0.16	0.2
Wall loops	3	2
Sparse infill density (%)	40	15
Sparse infill pattern	Gyroid	Gyroid
Travel speed (mm/s)	500	600
Support	Tree Slim	Tree Slim

5) COMPONENT MANUFACTURING

The manufacturing of prosthesis components considers the additive manufacturing process and the generation of electronic boards. First, the generation of force transmission elements and the casing is carried out using additive manufacturing. All 3D-printed parts of the developed prosthesis were fabricated using the "Bambu Lab X1 Carbon" machine from Bambu Lab (Texas, USA). The material used for these parts was polylactic acid (PLA), extruded through a 0.4 mm nozzle. The manufacturing parameters were determined based on the function of each component: those intended to withstand loads or transmit forces for mechanism activation and those serving as casing or protective covers. Table 2 presents the selected properties according to their classification.

The fabricated control electronic board also features a double-layer design with metalized pads. All components are assembled using surface-mount technology. Furthermore, the board includes mounting holes designed for secure attachment to the back of the prosthesis, ensuring stability and preventing relative displacement during operation. An assembly of the prosthesis and a view of the internal components can be seen in Fig. 5.

B. SOFTWARE

1) AI ALGORITHM PERSONALIZATION

According to Garcia et al. [31], XGBoost and Random Forest classifiers provide the best results in classifications (up to $93.4 \pm 2.4\%$) for different users with trans-radial amputations. These AI models were also tested on an embedded system (Raspberry Pi Zero 2W) and could generate predictions in real-time. Still, it also reports that the performance of the Random Forest classifiers on the embedded system was slower. Therefore, for this implementation, it was decided to follow their XGBoost approach to obtain personalized AI models for each user, which can be integrated into an embedded system. These models include multiclass XGBoost classifiers and also two-layer XGBoost stacked models, where the first layer classifies between rest and movement.

In contrast, the second layer is used when a movement is detected and classifies it as either flexion or extension. These models receive as input a segment of EMG signals with a window size set between 50 ms and 200 ms, which is considered as one of the model's parameters. Then, a feature extraction process is performed to obtain the Integrated EMG (IEMG), Modified Mean Absolute Value (MMAV), Root

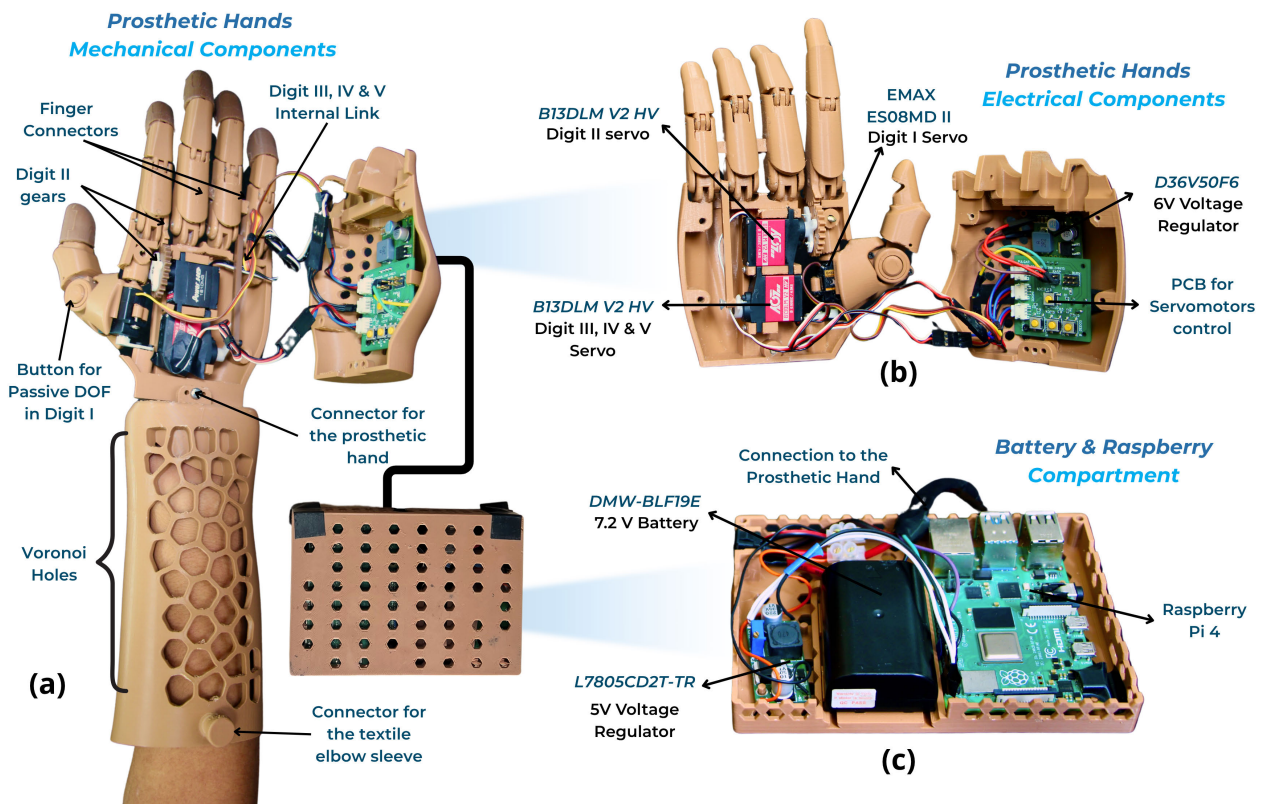


FIGURE 5. 3D-Printed prosthetic hand and socket fully assembled parts. (a) mechanical components, (b) electrical components, and (c) battery and Raspberry Pi compartment electrical components.

Mean Square (RMS), V-Order (V3), and Waveform Length (WL), which were found to be the most significant in the training process.

However, when users wore the prosthesis and tested these models, the algorithm presented a high rate of misclassifications. It was noted that the EMG signals captured while wearing the prosthesis differed significantly from those used for algorithm training, which were obtained without a prosthesis.

In order to improve the classification process, the gathering of EMG signals was performed again, this time with each user wearing their personalized prosthesis. Then, AI models were trained using the new data collected for each participant, as described by Garcia et al. [31]. After the models are trained for each user, they are stored for implementation on the Raspberry Pi 4.

After a model is loaded into the Raspberry Pi 4, to minimize involuntary activations and potential misclassifications, the algorithm requires consecutive identical classifications to confirm a movement, ensuring smooth and reliable transitions between gestures. A buffer system stores the last 5 to 15 classifications to improve accuracy, determining the final output based on the most frequent intent. A larger buffer enhances accuracy but requires more effort, while a smaller buffer allows quicker transitions with slightly reduced precision. The buffer size is adjustable, with an estimated

response time of 0.25 to 0.30 seconds, depending on its configuration.

For each user, model and parameter tuning are performed in real-time to achieve the best performance when using the prosthesis. First, the models are ranked according to their accuracy performance in training and are loaded onto the Raspberry Pi 4. Then, the model predictions are monitored while the user performs flexion and extension movements. If the readings differ from the movement performed, the model is updated. Afterwards, the buffer size is set. This parameter is adjusted based on the user's comfort, primarily because a larger buffer size requires the user to maintain their movement intention, which can cause fatigue in some cases. This process is repeated until the user is comfortable activating the prosthesis movements.

A complete overview of the model training and the algorithm's parameters tuning is shown in Fig. 6.

2) STATE-BASED GESTURE CONTROL

To enable intuitive and efficient control, the prosthesis operates using a state machine that processes the user's residual limb muscle activation as inputs to transition between different hand gestures, as seen in Fig. 7.

The system consists of six states: one passive gesture (the default extended hand) and five active gestures (Clamp, Cylinder 1, Cylinder 2, Handle, and Typing). Each active

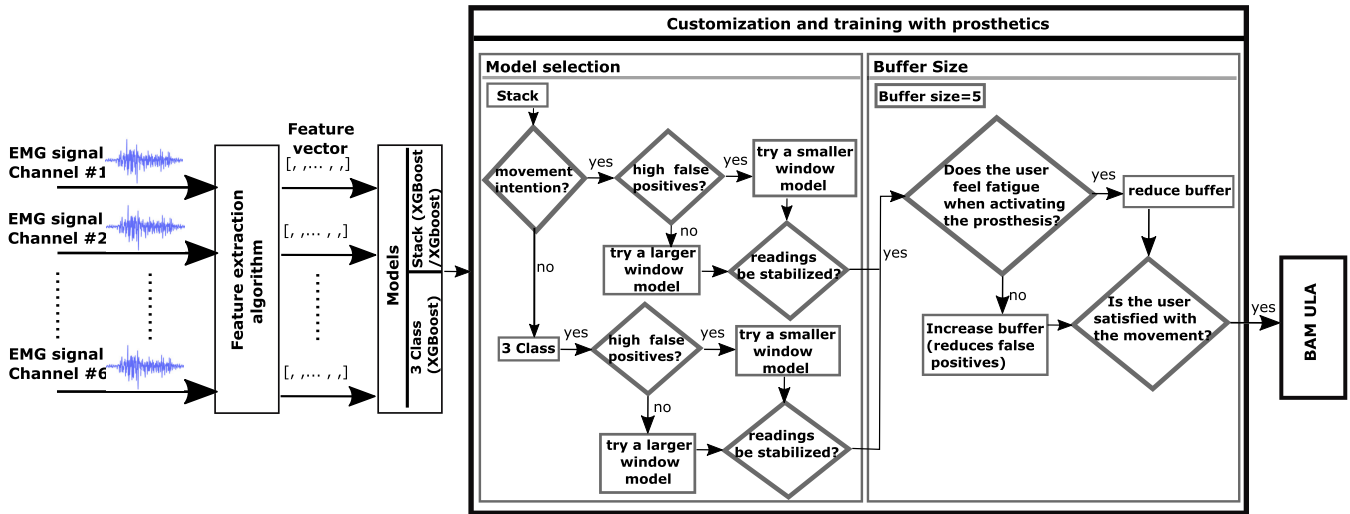


FIGURE 6. Complete algorithm selection flow diagram. Left: Model training using EMG signals. Right: Algorithm's parameter tuning while using the prosthesis.

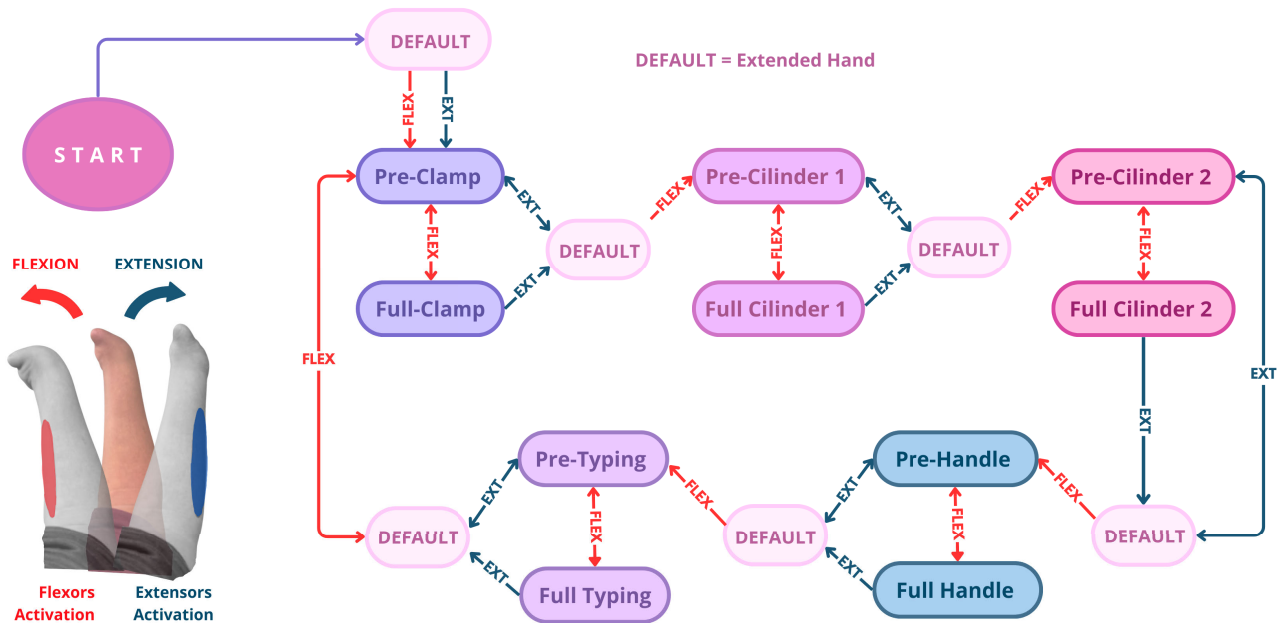


FIGURE 7. State machine for prosthetic hand control. The system transitions between a default state (Extended Hand) and five active gestures (Clamp, Cylinder 1, Cylinder 2, Handle, and Typing). Flexion movements switch from resting to active states, while extension movements return the hand to the resting position or transition to the next gesture.

gesture has two phases: a passive form, where the prosthetic thumb positions itself for the movement, and an active form, where the fingers close to complete the action. Flexion moves the system from passive to active within the same gesture, while extension resets the hand to the open position, allowing a transition to another gesture [30].

C. BAM-ULA ASSESSMENT PROTOCOL

The BAM-ULA protocol [32] is a clinically validated instrument that objectively measures upper-limb function in individuals using a prosthetic hand. It consists of ten standardized tasks requiring participants to manipulate objects or perform daily activities. Performance is scored on a binary scale, where successful task completion is awarded

one point, and failure is scored as zero. The total score, ranging from 0 to 10, quantitatively measures the prosthesis' functionality. Unlike other assessment protocols, BAM-ULA does not account for execution speed, number of attempts, or task complexity, focusing solely on functional feasibility.

The activities are listed as follows:

- **BAM-ULA 1:** Tuck in a shirt by stabilizing the pants, inserting the shirt evenly, and securing it within the waistband.
- **BAM-ULA 2:** Lift a bag using a secure grip and place it on a shelf at eye level with controlled movements.
- **BAM-ULA 3:** Open a water bottle with the intact hand, drink using the prosthetic side, and place it back with control.

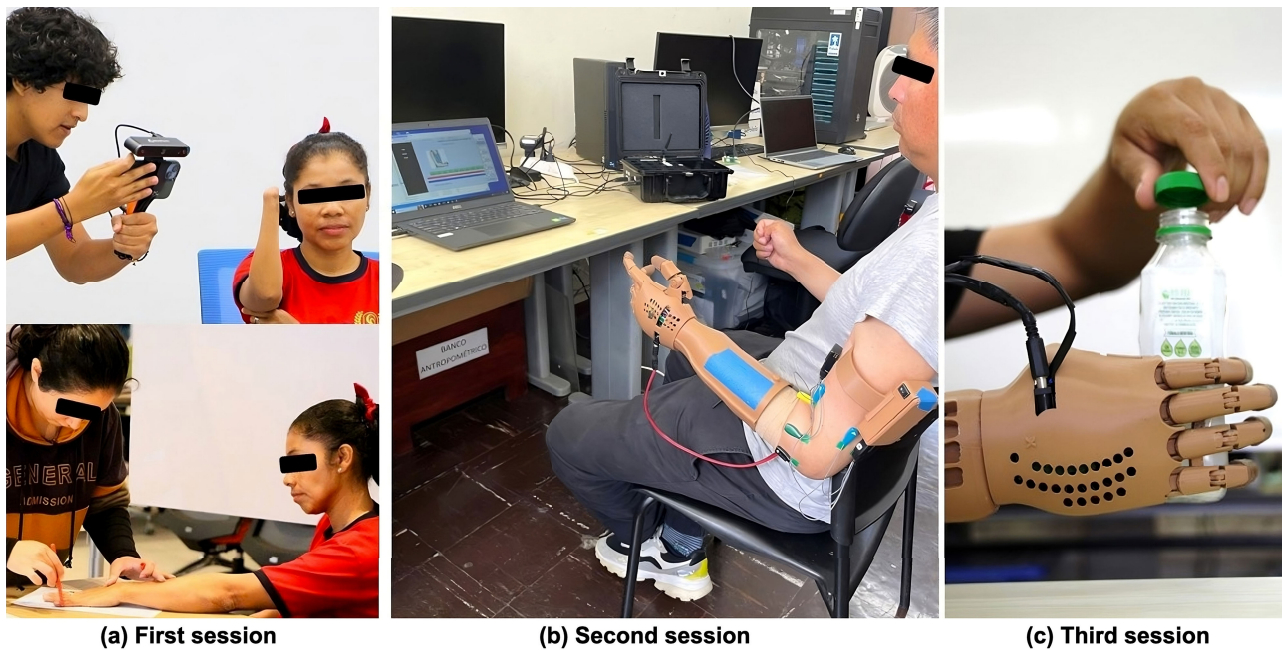


FIGURE 8. Sequential protocol for prosthesis fitting, AI training, and functional assessment. (a) 3D scanning and anthropometric measurements of the residual limb and the contralateral hand, (b) initial prosthetic socket fitting and sEMG data acquisition for AI-based control training, with the volunteer executing predefined movements while wearing the prosthesis, (c) familiarization with prosthesis use and functional assessment using the BAM-ULA protocol.

- **BAM-ULA 4:** Retrieve a wallet from the back pocket using the prosthetic side with proper grasping and extraction.
- **BAM-ULA 5:** Insert a wallet into the back pocket using the prosthetic side, ensuring controlled placement and release.
- **BAM-ULA 6:** Remove a plastic jug from the fridge and place it on a countertop using the prosthetic side.
- **BAM-ULA 7:** Open a jug, pour liquid into a cup, and return the jug upright using the prosthetic side.
- **BAM-ULA 8:** Brush or comb hair using the prosthetic side, ensuring controlled motion and grip.
- **BAM-ULA 9:** Use a fork with the prosthetic side, bringing it to the mouth and placing it back on the table.
- **BAM-ULA 10:** Open a door using the prosthetic side by grasping, rotating the handle, and releasing.

D. EXPERIMENTAL PROCEDURE

1) PARTICIPANTS

The study included nine adult participants aged 18 to 60 years with unilateral transradial amputations due to traumatic causes. Inclusion criteria required participants to have a residual limb length of at least 10 cm, no comorbidities, and literacy to follow instructions. Exclusion criteria ruled out those with bilateral amputations, comorbid conditions, or severe muscle atrophy.

Participants were recruited through a public call disseminated via social media and in coordination with rehabilitation centers. All participants underwent an evaluation by a

licensed occupational therapist to validate eligibility before providing informed consent.

2) TESTING PROTOCOL AND PERFORMANCE EVALUATION

For this study, nine fully personalized prosthetic sockets were fabricated to match each participant's residual limb anatomy and to ensure precise positioning of the sEMG electrodes. To optimize resources while maintaining ergonomic functionality, a modular approach was adopted for the prosthetic hand units. Specifically, four prosthetic hands were manufactured: two left-hand models (small and regular sizes) and two right-hand models (small and regular sizes). All nine participants' contralateral hand dimensions were within the 40th to 60th percentile range based on anthropometric reference data [33]. Participants with hand sizes in the 40th to 50th percentile range were assigned the small-size hand model, while those within the 50th to 60th percentile range used the regular-size version. The sockets and prosthetic hands were mechanically linked via a distal universal connector, enabling seamless interchangeability between components.

This study was conducted across three sessions, as shown in Fig. 8. During the first session, a 3D scanning procedure of the amputated limb segment and anthropometric measurements of the contralateral limb were performed. This procedure initiated the design process for the customized prosthetic hand and its tailored socket, ensuring an optimal anatomical fit for the participant. During the second session of the study, participants fitted and tested their personalized prosthetic sockets. After the testing, an EMG data capture

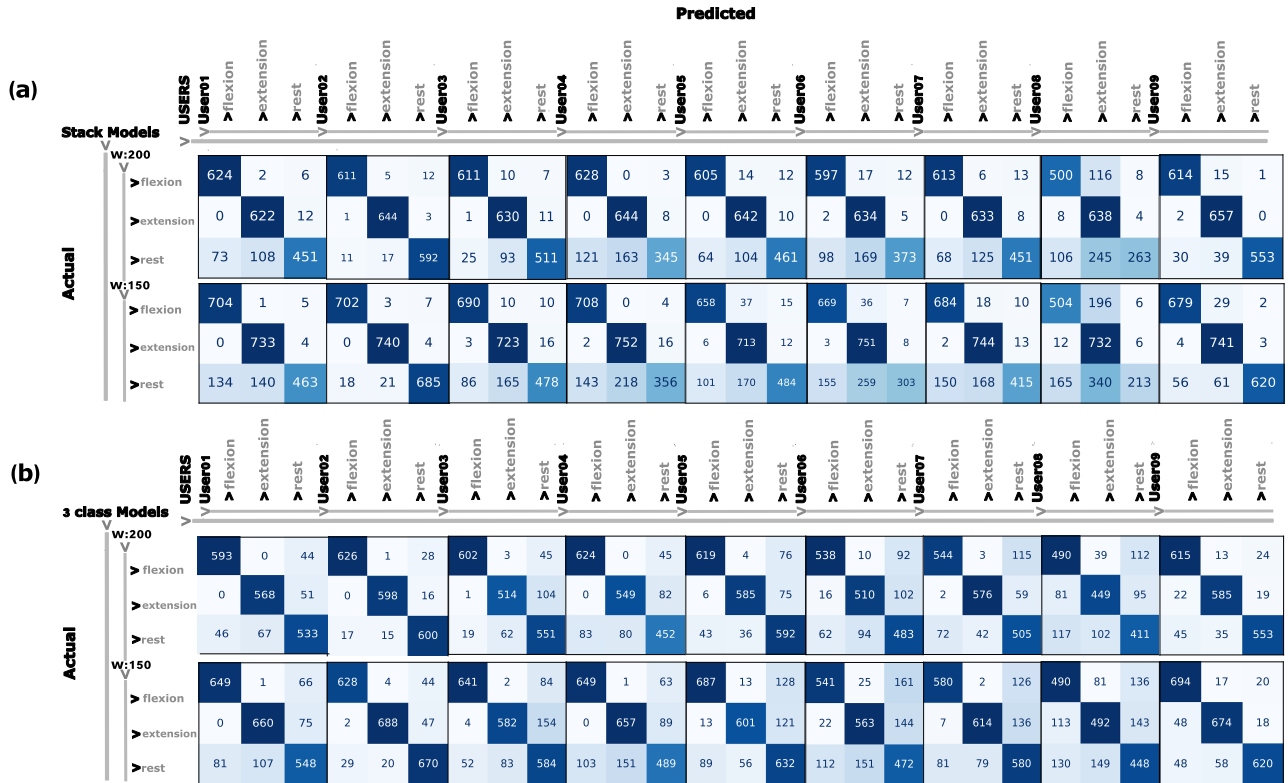


FIGURE 9. Confusion matrices for each user for the XGBoost trained models, using a window size of 150ms and 200ms, (a) models using the two-layer stacked architecture, (b) a multiclass classification approach targeting three distinct classes.

TABLE 3. Accuracy metrics for the trained models.

Window Size	multiclass				two-layer stack			
	50	100	150	200	50	100	150	200
User 1	75.2	80.4	84.9	89.1	75.8	82.3	87.0	89.4
User 2	89.3	93.1	93.3	95.9	93.1	96.1	97.6	97.4
User 3	71.7	76.7	82.7	87.7	74.6	82.4	86.7	92.3
User 4	71.9	76.7	81.5	84.9	70.8	77.2	82.6	84.6
User 5	67.5	74.8	82.1	88.2	68.2	78.0	84.5	89.3
User 6	59.5	67.2	71.9	80.3	62.0	71.9	78.6	84.1
User 7	69.1	73.9	80.5	84.7	71.3	79.2	83.6	88.5
User 8	50.5	62.7	65.5	71.2	52.3	64.5	66.7	74.2
User 9	79.1	86.9	90.5	91.7	81.6	89.1	92.9	95.4

session was performed using the sEMGUI software [34] while each participant wore their prosthesis. Finally, during session 3, the participant underwent a familiarization phase in the first half. During the familiarization phase, participants practiced the opening and closing of the prosthetic hand while transferring four types of objects: (1) extensive and lightweight, (2) large and medium-weight, (3) small and lightweight, and (4) small and medium-weight. Lightweight objects weighed no more than 100 g, while medium-weight objects did not exceed 400 g. As appropriate, participants were instructed to grasp each object with their intact hand and transfer it to the prosthetic hand using a cylindrical or pinch grip. They completed ten repetitions for each object type. Subsequently, they reversed the task, picking up the object with the prosthetic hand and transferring it to the intact

hand. After the familiarization was completed, the participant performed the BAM-ULA assessment protocol. Key metrics were collected to evaluate the prosthesis's performance during this assessment. These included the completion time for each task, binary task scores (0 or 1) as detailed by the BAM-ULA protocol, qualitative usability observations of participant interaction with the prosthesis, and battery consumption during task execution.

III. RESULTS

A. AI MODEL TRAINING

The classifier algorithms to be implemented in the prosthesis were trained with the new dataset obtained in this study. Table 3 shows the accuracy obtained for each of the trained models. To further interpret the results of this training, Fig. 9 shows the confusion matrices for the best models for each user, separating the multiclass model and the two-layer stacked model.

As can be seen in the confusion matrices, most flexion/extension movements are correctly predicted; however, the rest stance has a significant number of misclassifications, with some users experiencing nearly 50% incorrect predictions. This results in a decrease in the overall accuracy of the best models, which ranges between 84.1% and 97.4%, excluding the lowest value. Regarding this value, user 8 had the shortest amputation level, which results in smaller muscular groups and also leads to having the EMG



FIGURE 10. Activities of the BAM-ULA assessment. (a) lifting a bag, (b) brushing or combing hair using the prosthetic side, (c) opening a jug, (d) using a fork with the prosthetic side, and (e) drinking using the prosthetic side.

sensors closer to the elbow. Despite this problem, most of the misclassifications ended up appearing on the rest stance, so the models could still be used for our purpose.

B. BAM-ULA ASSESSMENT

Upon completing the assessment protocol, total scores were computed for each participant, with a maximum possible score of 10 points.

There were nine subjects with a mean age of 35.9 years (standard deviation (SD) 8.7 years), 56% male. Regarding amputation levels, 67% had distal amputations, and 33% had middle-level amputations. Participants had a mean duration of 23.8 years without a limb (SD 11.3 years). The majority had had amputations due to accidents (66.7%). They were predominantly prosthesis users, with 67% having prior experience: 11% using a myoelectric prosthesis and 67% using a mechanical device. Fig. 10 shows participants performing some assessment tasks. Table 4 shows how many participants achieved each of the tasks asked by the BAM-ULA protocol. Regarding user experience, participants found the prosthesis easy to use, especially with more significant movements. Those unfamiliar with myoelectric control needed more time to adjust. Some users reported minor discomfort after extended use because their skin tended to sweat due to personal conditions related to their amputation. For volunteers without prior experience using prostheses, longer pauses were observed between successive task attempts in the protocol.

TABLE 4. BAM-ULA Scores.

Tasks	N(%)
1. Tuck shirt	8 (88.9)
2. Bag to shelf	9 (100.0)
3. Water bottle	9 (100.0)
4. Remove wallet	2 (22.2)
5. Replace wallet	4 (44.4)
6. Lift gallon jug	9 (100.0)
7. Open and pour jug	8 (88.9)
8. Brush/Comb hair	7 (77.8)
9. Use a fork	9 (100.0)
10. Open door with knob	2 (22.2)
	Mean (SD)
Overall performance rating	7.4 (0.7)

SD: Standard deviation.

They reported experiencing a slight increase in muscle fatigue, as they were not accustomed to performing flexion and extension attempts.

Considering performance variability among participants, scores varied based on experience. Participants who had used a prosthesis before generally scored higher, while those using one for the first time had lower scores. The lowest score recorded was 6/10, while experienced users achieved a score of 8/10. Those with middle-level amputations had slightly more difficulty with precise movements compared to those with distal amputations.

The BAM-ULA assessment results revealed slight differences in performance between participants with congenital and accidental amputations. Participants with accidental



FIGURE 11. 3D-Printed modular sockets for upper-limb prostheses: customizable designs for enhanced fit and ventilation.

amputations had an average score of 7.5/10, while those with congenital amputations averaged 7.33/10. This suggests a marginally higher functional performance among individuals with accidental amputations.

Regarding the influence of years without a limb on performance, participants in this study had lived without a limb for an average of 23.8 years (SD = 11.3). The range was broad, with the shortest duration being 3 years and the longest reaching 35 years.

To explore potential trends, participants were divided into short/mid-term (0–28 years) and long-term (>28 years) amputees. Short/mid-term amputees, representing five participants, had an average BAM-ULA score of 7.5/10, while the four long-term amputees had an average score of 7.3/10.

The results suggest that short/mid-term amputees performed slightly better on average. This could indicate that recent adaptation efforts or rehabilitation experiences have contributed to their performance. However, the difference is minimal, reinforcing that prosthetic adaptability remains effective regardless of the years since amputation.

IV. DISCUSSION

A. REGARDING HARDWARE

Considering the anatomical customization through digital manufacturing, the prosthetic socket was designed to replicate the natural shape of the arm, incorporating a 1.2 mm offset between the skin and the inner socket layer to ensure comfort during wrist flexion. This approach considers anatomical reference points and aligns them along the axial axis, ensuring a symmetrical fit. A significant limitation of standardized sockets is the lack of personalization, which can lead to instability or discomfort. If the socket is too tight, it can cause discomfort. In contrast, an oversized socket requires additional retention methods, such as self-adhesive bandages or Velcro straps, which can increase system weight and potentially cause heat buildup. The socket's curved trajectory and thickness are derived from scan data, though manual adjustments may be necessary for irregular residual

limb geometries. Fig. 11 shows some variants of developed prostheses sockets. Material constraints also influence usability [35], as variations in thickness due to manual adjustments can introduce localized structural weaknesses or increase weight.

In this regard, evaluating the impact of additive manufacturing on the prosthesis weight and its ergonomics, the developed prosthesis, including its power system, weighs 493 g, which falls within the range of comparable prostheses such as the HeroArm [36] (340 g) and others such as UnlimitedTomorrow [37] (370 g) and DEKA arm [38] (583 g). Initial trials suggest that weight did not significantly impact usability; however, prolonged use revealed minor deformations in the socket-hand interface and internal gears. High-strength materials such as Acrylonitrile Butadiene Styrene (ABS) or Acrylonitrile Styrene Acrylate (ASA) could be considered in future iterations to enhance durability.

Regarding its functional design and joint movements, Recent studies on EMG-actuated prostheses utilize actuators similar to those in the developed prosthesis. Compared to [12], which features a 2-DOF prosthetic hand driven by a single digital servomotor with two grasping patterns, and [39], [40], which employs five micro DC motors to achieve 5–8 grasping patterns, the approach proposed in this paper offers a balance between the number of actuators and the variety of grasping gestures evaluated during user trials. Additionally, compared to the DS3225 servomotor, which provides 21–25 kgf-cm, and the brushless DC motor, which delivers 0.3–0.4 kgf-cm, the proposed design incorporates the Power HD 1810MG servo, rated at 3.9 kgf-cm.

In terms of prosthetic socket and user experience, commercial prostheses, such as [40] and [41], integrate EMG sensor placements directly within the socket design, simplifying prosthesis fitting and eliminating the need for technical expertise in electrode positioning. In contrast, traditional high-end prostheses rely on suction or mechanical locking systems [42], which provide a stable fit but require precise adjustments and can create high localized pressure, leading to discomfort over time. The developed bespoke prosthetic socket incorporates an adjustable textile elbow sleeve that interfaces the rigid socket and the user's skin, offering a more adaptable and comfortable fit by distributing pressure more evenly, as shown in Fig. 12. Unlike suction and mechanical locking systems, which depend on an airtight seal or mechanical engagement, this design passively secures the prosthesis while also enclosing and protecting the EMG electrodes and wiring, such as the Trigno® Quattro Sensor 2024 units used in this study, manufactured by Delsys Incorporated. Although the textile sleeve may shift slightly during use and require occasional readjustment, it facilitates easier donning and doffing while reducing the risk of skin irritation commonly associated with rigid socket materials.

The battery system supports continuous operation for 4 to 6 hours, with power consumption primarily driven by

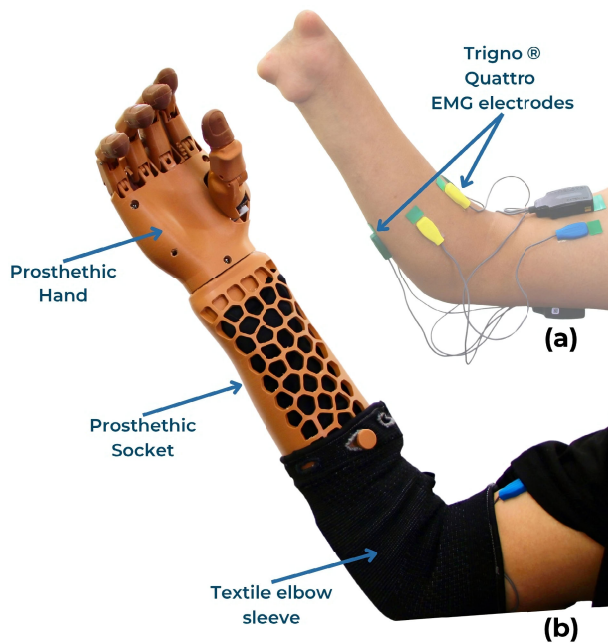


FIGURE 12. Sensor placement and adjustable elbow textile. (a) shows the amputated section with the positioned electrodes, (b) displays the adjustable textile elbow sleeve and its arrangement in the prosthetic socket.

real-time AI-based control. Algorithmic optimizations have been implemented to reduce computational load; however, temporarily pausing signal processing could also extend battery life. Lightweight battery modules enable quick replacements, thereby mitigating downtime.

Future iterations should explore energy-efficient AI models and adaptive power management strategies to enhance system efficiency further.

Environmental factors, including lighting conditions and surface textures, impact the accuracy of 3D scanning used for socket fabrication. Irregular residual limb geometries introduce additional challenges, requiring post-processing refinements. Compact internal component design complicates maintenance, particularly in securing motor assemblies. Some instances of prosthesis slippage were addressed through adjustments to textile fastening, although further refinements in socket integration and power system wiring management could enhance overall usability.

B. REGARDING SOFTWARE

The AI-based control system in the transradial myoelectric prosthesis processes sEMG signals using pattern recognition models to improve gesture classification accuracy. Unlike fixed-threshold methods, which rely on predefined amplitude values, the AI model adapts to variations in muscle activation patterns and shifts in electrode placement, thereby reducing classification errors. Future improvements include low-power AI inference models and hybrid control strategies integrating rule-based heuristics to enhance energy efficiency and response time while maintaining classification accuracy [43].

The EMG sensor used in this study features an integrated band-pass filter, which effectively enhances EMG signal quality without requiring additional processing. This built-in filtering significantly reduces the computational load on the control system, eliminating the need for an independent band-pass filter between 20 Hz and 500 Hz. Typically, such a filter is necessary to remove low-frequency noise (below 20 Hz) caused by electrode movement, cable displacement, or general equipment motion. Additionally, EMG data within the 20 Hz to 100 Hz range improves signal accuracy for transradial amputees by 1.6%. In comparison, 95% of the dominant signal energy is concentrated in harmonics between 400 Hz and 500 Hz [44]. A low-pass filter at 1000 Hz is commonly used, but lowering the cut-off frequency to 500 Hz reduces processing time and memory usage with minimal loss of relevant information. Further improvements in signal quality could be achieved by eliminating the 60 Hz frequency component, which removes noise generated by power line interference.

The prosthetic operates using a state machine that classifies muscle activity as flexion, extension, or rest through an artificial intelligence algorithm. Consecutive identical classifications confirm movement intent, reducing involuntary activations and ensuring smooth transitions between gestures. A buffer mechanism stores between 5 and 15 previous classifications to determine the most frequent intent. This helps alleviate the misclassifications that occur during a rest stance, which leads to false activations on the prosthesis. Larger buffers improve accuracy but require more muscle effort from the user, whereas smaller buffers allow quicker transitions with slightly reduced precision.

The wireless EMG sensors required proximity to the central processing module to ensure stable signal transmission. No interference issues were observed during sensor operation. However, a wired connection was established between the Raspberry Pi and the PCB, eliminating the need for Bluetooth or Wi-Fi communication. This decision was made due to limited space within the prosthetic housing, the close placement of electronic components, and the need to avoid potential communication failures caused by servo motor operation or electromagnetic interference from nearby electrical components. During testing, delays and system freezing were observed when EMG data packets exceeded 100 samples. To address this, parallel processing was implemented, allowing for the simultaneous acquisition of EMG signals and updating the data list, thereby preventing processing delays. However, this approach increased computational load, requiring system resources to be managed efficiently. A serial processing approach was initially considered but discarded due to its slow response times, which negatively affected real-time user interaction with the prosthesis.

The continuous operation of the AI system is the primary factor contributing to the highest energy consumption, as it prioritizes faster response times over power efficiency.

A lightweight operating system was implemented on the Raspberry Pi to optimize power usage, reducing continuous power consumption by 50 mA without compromising system performance. Additionally, future improvements could include a preprocessing stage designed to detect signal peaks or abrupt changes in the EMG data before the AI algorithm is engaged. This approach would enable the selective activation of AI processing, thereby reducing overall computational demand and extending battery life while maintaining real-time responsiveness.

The user interface and customization of the prosthetic control system are designed to generate a user-specific model rather than employing a generalized adaptive learning framework. Each user undergoes a data collection phase, allowing the system to develop a model that adapts to their specific muscle activation patterns. This approach ensures the prosthesis operates based on individual signal characteristics, improving gesture recognition accuracy and control consistency. Although the system was trained using standard gestures such as flexion and extension, these movements can be adjusted or extrapolated based on the user's capabilities. The personalized model is not restricted to predefined motions but can incorporate any two distinguishable gestures that the user can reliably perform. This flexibility enables better adaptation to individual motor abilities, addressing residual limb conditions and variations in neuromuscular control. Future work could explore expanding this customization process, enabling users to define their gesture set within a structured calibration phase. Additionally, implementing dynamic model updates based on long-term user data could enhance the prosthesis's adaptability, further optimizing control performance over time.

The post-processing stage, particularly the buffering mechanism, introduces a slight delay in response time in exchange for higher classification accuracy, while remaining within the required operational limits. The feasibility of offloading AI processing to a mobile device or cloud-based system was explored to optimize computational efficiency, with a focus on ensuring reliability and low-latency performance. Improving the classifier algorithm also improved post-processing efficiency, ultimately reducing overall processing time. Observations indicated that a buffer size of 3 to 5 readings was sufficient for some users to maintain stable performance in flexion/rest predictions. However, it was observed that the algorithm had problems in predicting extension movements, despite achieving a high identification rate during the training process. This could have been produced because there was at least a week between the data collection and the prosthesis tests. Resnik et al. [25] state that prosthesis users lost dexterity while handling their myoelectric prosthesis after one week of training the pattern recognition algorithm, which implies that users have to undergo constant training to prompt the corresponding algorithm correctly. Additionally, as users make movements

while holding a variable weight, this can affect EMG readings and requires further analysis [45].

Considering the AI processing times, the measured system response time ranged from 0.1 to 0.2 seconds, ensuring real-time operation while balancing computational requirements.

Regarding the trade-offs between AI's efficiency and autonomy, a key advantage of the implemented AI model is that it eliminates the need for manual tuning. Instead of manually adjusting parameters for each user, the system follows a standardized pipeline, collecting user-specific data and executing a training algorithm that generates a personalized model within 10 to 15 minutes. This process ensures optimized control with minimal intervention. The trained model is directly deployed on the Raspberry Pi 4, enabling on-device gesture classification without external processing. This aligns with the low-cost design strategy, enabling local model training and deployment on embedded hardware, which significantly enhances accessibility and practicality in resource-constrained environments.

C. REGARDING PROSTHESIS ASSESSMENT

The evaluation of our transradial myoelectric prosthesis using the BAM-ULA protocol yielded an average score of 7.4 out of 10 across nine participants, with a standard deviation of 0.7, indicating consistent performance among users and minimal variability in task success. Multiple participants achieved the highest scores (8/10), while one obtained the lowest (6/10). Gross motor tasks, such as placing a bag on a shelf (BAM-ULA 2), lifting a gallon jug (BAM-ULA 6), and using a fork (BAM-ULA 9), were completed successfully by all participants (100%), confirming that the prosthesis provides stable grip force and adequate weight support. In contrast, fine motor tasks, such as retrieving a wallet from a back pocket (BAM-ULA 4) and opening a door (BAM-ULA 10), had the lowest success rates (22.2%), highlighting current limitations in precision grip and wrist articulation.

The accessibility of the proposed prosthesis is significantly enhanced by its cost-effective production, ease of acquisition, and simplified reparability, which are crucial factors for long-term user satisfaction. Unlike high-end commercial devices, which are often prohibitively expensive and difficult to maintain in low-resource settings, this prosthesis utilizes additive manufacturing and modular components, enabling affordable customization and straightforward repairs. Users benefit from a system that can be locally manufactured and serviced, reducing downtime and ensuring continued usability. This accessibility improves adoption rates and fosters greater independence and long-term engagement, as users are more likely to integrate the device into their daily lives without financial or logistical barriers.

A comprehensive bill of materials, encompassing both mechanical and electronic components, indicates a production cost of approximately USD 780 per prosthetic unit. This excludes the Trigno[®] Quattro EMG sensors by Delsys

Incorporated, which were used in this study and are typically reserved for research due to their high cost and advanced signal acquisition capabilities. These sensors require wireless communication with a Raspberry Pi unit, as shown in Figure 4, and are not intended for large-scale deployment. As the system progresses toward higher technological readiness levels (TRLs), more affordable and commercially viable EMG acquisition sensors will be integrated to enable real-world testing and autonomous daily use.

When comparing these results to the DEKA Arm, a commercially available advanced myoelectric prosthesis evaluated under the same BAM-ULA protocol, differences emerge [46]. In the DEKA study, transradial participants achieved an average initial BAM-ULA score of 6.3 out of 10 ($SD = 2.8$), which is lower than the initial performance of our prosthesis (7.4/10, $SD = 0.7$). This suggests that our device provided a more consistent and intuitive user experience during early testing. However, after a home-use phase, during which participants incorporated the DEKA Arm into their daily life over several weeks, their average score improved to 8.3/10, reinforcing the importance of long-term adaptation and real-world exposure.

Task-specific trends show that both prostheses excel in gross motor tasks, but fine motor tasks remain the most challenging. However, one key difference was observed: while only 22.2% of our participants successfully opened a door with a knob (BAM-ULA 10), the DEKA Arm achieved a 68.6% success rate for this task. This suggests that wrist articulation is critical in door-opening performance and should be a focus for future design refinements. Despite this difference, our prosthesis demonstrated higher early-stage usability, which may still affect the early-stage usability of BAM-ULA scores more than that of the DEKA Arm, reinforcing its potential as a functionally competitive and cost-effective alternative.

While both prostheses share similar BAM-ULA scores, they also exhibit common limitations. The DEKA Arm demonstrated slower task execution, with the Timed Measure of Activity Performance (T-MAP) showing an effect size (ES) of 0.76, indicating moderate delays in laboratory testing. However, performance improved with extended home use, highlighting the need for long-term adaptation. Similarly, the developed prosthesis, despite its AI-driven personalized control, requires an initial calibration phase to generate individualized models of muscle activation. However, muscle signal variability and learning requirements may still affect age usability. Another shared limitation is that both prostheses also face high energy consumption. The DEKA Arm's multiple actuators increase power demands, leading to frequent recharging requirements. While the developed prosthesis incorporates low-power AI inference, battery life remains a concern due to the continuous processing of EMG signals and motor actuation.

Several refinements should be considered to enhance functional performance further and close the gap with

the DEKA Arm's post-home-use results. First, improving the grip mechanism and incorporating wrist articulation adjustments could enhance precision control, particularly in retrieval task and door-opening tasks. Additionally, some participants encountered difficulties maintaining the prosthetic hand at face level or higher, particularly when attempting to brush or comb hair (BAM-ULA 8). This suggests that the current gesture set for EMG control does not fully capture muscle activation patterns for elevated arm postures. Expanding the machine learning training dataset to include EMG signals from a broader range of shoulder and arm positions could improve the system's ability to recognize and execute these movements. For middle-level amputees, introducing a wrist articulation unit could provide greater control in pronation, supination, and wrist flexion, which are currently limited by the rigid wrist design. This would likely improve fine motor and rotational task performance, thereby increasing task success rates for activities that require precise wrist positioning.

Other commercially available unilateral myoelectric prostheses for transradial users with multiple active degrees of freedom (multi-DOF) have also been evaluated using the BAM-ULA instrument. In a 2020 study by Resnik et al. [47], a cohort of 127 U.S. military veterans with upper limb amputations was assessed to examine dexterity, activity performance, disability, quality of life, and independence among users of prostheses. Among them, 18 participants used transradial multi-DOF myoelectric prostheses, which are comparable in control architecture to the device presented in this study. Their performance was evaluated across multiple outcome measures, including BAM-ULA scores. According to the reported results, these 18 participants achieved an average BAM-ULA score of 8.0 ($SD = 1.6$).

Although the study by Resnik et al. does not specify the number of users fitted with each prosthetic type, it reports that the following commercial devices were used: i-Limb [48], Michelangelo [49], Bebionic [50], and LUKE Arm [51].

While the score reported in Resnik et al. is slightly higher than that obtained by participants using the prosthesis developed in this study, it is essential to note that the 18 individuals in their cohort were assessed while using their own prosthesis: Devices that had already been integrated into their daily routines. In contrast, the 9 participants in this study were evaluated after a brief period of initial exposure to the device.

Therefore, the 8.0 score obtained in the Resnik study is more appropriate than the post-home-use score of 8.3 reported for the DEKA Arm, as discussed in this subsection. Both reflect performance after sustained daily use. This highlights the significant role of user adaptations and long-term integration in improving functional outcomes.

Table 5 presents a comparison between the prosthesis developed in this study and the other devices discussed in this subsection.

TABLE 5. Comparison between the developed prosthesis and state-of-the-art commercial prostheses.

Feature	Study Prosthesis	DEKA Arm	i-limb	Michel- angelo	Bebionic	LUKE- Arm
BAM-ULA Score (Initial Tests)	7.4/10	6.3/10	N/A	N/A	N/A	N/A
BAM-ULA Score (Long-Term Use)	Not tested yet	8.3/10	8/10	8/10	8/10	8/10
Control Mechanism	AI-based, user-personalized	Pre-set control algorithms	Myoelectric with gesture switching	Myoelectric with an intelligent thumb	Proportional control	Myoelectric, IMU sensors
Customization	Parametric, bespoke socket	Standardized commercial sizing	Custom socket, mobile app configuration	Anatomical design, AxonBus system	Standard commercial sizes	Modular design
Dexterity	6 predefined gestures	Multi-grip configurations	up to 24 programmable grips	7 grip types	14 grip patterns, individual finger control	Multiple, proportional speed control
Wrist Articulation	Fixed	Articulated wrist for rotation	Optional powered wrist rotator	AxonWrist: flexion-extension, rotation	Multiple wrist configuration	Active Wrist, 2 DOF
Power Consumption	2100 mAh, 7.2V	5000 mAh, 9 V	2 options: 1300 and 2000 mAh, 7.4V	1150 mAh, 11.1V	1300 mAh, 7.4 V	5000 mAh, 14.8V
Low-Resource Availability	High	Low	Low	Low	Low	Low

While the prosthesis demonstrated promising performance in a controlled environment, real-world deployment presents additional challenges that require further study. Factors such as long-term usability, environmental variations, and user adaptation must be considered to ensure consistent functionality in daily activities. Unlike structured laboratory tests, real-life use introduces unpredictable variables, including temperature fluctuations and humidity that affect EMG sensor performance, as well as variations in user engagement over time.

To address these concerns, longitudinal home-use studies should observe how users integrate the prosthesis into their routines, how EMG signal stability evolves with prolonged use, and whether comfort or usability degrades over time. Additionally, battery consumption patterns during extended daily use must be analyzed to optimize power efficiency.

Comparisons between laboratory-based and real-world task performance would provide insight into how external factors affect the precision of AI-driven gesture recognition and control. Furthermore, assessing prosthesis acceptance in work and social environments would enable refinements that enhance practicality and promote long-term adoption.

V. CONCLUSION

Developing a personalized transradial myoelectric prosthesis, combining additive manufacturing and AI-based control, exemplifies how emerging technologies can converge to address critical challenges in prosthetic care. Beyond demonstrating technical feasibility, this work highlights the importance of user-centered design and adaptive control strategies in enhancing the functionality and user experience of upper-limb prostheses.

The proposed system overcomes the discomfort and instability often observed with standardized prostheses by tailoring the socket and hand geometry to each user's unique anatomy. Moreover, the AI-enhanced control system offers an adaptable interface that learns from individual muscle patterns, fostering more intuitive and natural interaction. These advancements contribute to improved performance in functional assessments, such as BAM-ULA, where the

prosthesis demonstrated an average score of 7.4 out of 10 across nine participants. This result indicates robust functional performance comparable to that of high-end commercial devices, reinforcing the prosthesis's viability as a competitive, cost-effective alternative.

Importantly, this work highlights how localized digital manufacturing can enable affordable customization, a key factor in increasing access in low-resource settings. This approach could be extended to other assistive technologies, fostering a broader personalized device ecosystem that addresses diverse rehabilitation needs.

Despite the favorable results, the development and implementation of the system presented several challenges. Functionally, isolated false-positive detections were observed during the evaluation stage. These were mitigated through a post-processing filter based on a configurable output buffer, which leverages the temporal continuity of valid classifications to suppress spurious signals. On the hardware side, the initial microprocessor exhibited excessive thermal output, prompting the selection of a higher-memory, low-thermal-footprint model from the same product family, preserving power compatibility while improving operational stability. Manufacturing challenges were addressed by iteratively optimizing the print orientation and slicing parameters of 3D-printed components, in accordance with the VDI 2221 product design methodology. From an ergonomic standpoint, variability in forearm perspiration across users was managed by incorporating Voronoi-patterned socket structures that enhance breathability. Institutionally, the absence of a national registry of upper-limb amputees in public healthcare settings limited the pool of eligible volunteers. To overcome this, previous participants were invited to participate again, and a targeted outreach campaign was conducted via social media.

Future research should explore the deployment of this technology in real-life environments, examining its long-term impact on quality of life, and integrating additional sensing modalities to enhance control robustness. Furthermore, the methodology presented here could inspire the development of new design frameworks for other medical devices, thereby

reinforcing the role of digital and intelligent technologies in driving accessible and personalized healthcare solutions.

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